

BracketNet: Learning Lie-Algebraic Structure from Unlabeled Transformation Triples

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Abstract

A latent space in which transformations act linearly does not guarantee that the learned infinitesimal generators form a Lie algebra. Generators can reconstruct local transitions while their commutators leave the learned span. The representation is then algebraically inconsistent, and it becomes unclear whether the transformation family is abelian. We study BracketNet, a transition-based representation learner that regularizes the closure of learned skew-symmetric generators under the matrix commutator. Training follows a fit–close–refit schedule. We show that zero closure yields a Lie subalgebra integrating to a connected matrix Lie group. We also bound the BCH escape from the learned span by $O(\delta r^2)$, where δ is controlled by the closure loss. On nonlinear observations of $\text{SO}(2)$, T^2 and $\text{SO}(3)$ actions embedded in a larger ambient $\mathfrak{so}(d)$, the closure weight turns out to be scale-sensitive. At the originally planned weight ($\lambda=0.1$) closure regularization had no detectable effect within the 420-update budget. A weight selected on validation seeds ($\lambda=30$) lowered mean closure residual relative to a composition-only learner by 98.9% on T^2 and 98.6% on $\text{SO}(3)$, on 5/5 held-out seeds each. The exact sign-flip $p = 0.0625$ is the minimum attainable with five seeds. With this weight, a commutator threshold fixed on validation classified all ten held-out abelian/non-abelian runs (baselines: 9/10). Independently trained $\text{SO}(3)$ models also recovered the *same* algebra up to latent conjugacy (subspace distance 0.034 vs. 0.42–0.45 for baselines). Linear CKA does not see this difference. Chained-transport differences were not statistically significant. We also report two limits. The closure loss is invariant only to orthogonal changes of generator coordinates. And exact closure does not identify the representation: one run converged to a chiral $\mathfrak{su}(2) \subset \mathfrak{so}(4)$.

1 Introduction

Symmetry is a central organizing principle for representation learning. When the group action is known, equivariant architectures can encode it directly [2, 1]. In many systems, however, the action is not available in observation coordinates. A growing line of work therefore learns symmetries or symmetric latent spaces from data [3, 11, 12, 7, 8, 4].

Transition-based objectives usually enforce local transport, two-step composition, or both. Neither requires that the learned infinitesimal directions close under commutation. For learned generators $A_1, \dots, A_K$ the missing condition is

$$[A_i, A_j] = A_i A_j - A_j A_i \in \text{span}\{A_1, \dots, A_K\} \quad \forall i, j. \quad (1)$$

Without (1), the nominal K -dimensional family need not be locally closed under multiplication. BracketNet penalizes the component of each commutator orthogonal to the learned span and uses a staged optimizer.

Contributions. (1) A differentiable Lie-closure objective, with a precise statement of its invariances (Sec. 4.1). (2) A fit–close–refit schedule. (3) A BCH bound in which escape from the span is $O(\delta r^2)$ with $\delta \leq \frac{K}{2} \sqrt{\mathcal{L}_{\text{close}}}$, plus a stability bound for chained orthogonal transport. (4) A benchmark in which the true algebra is embedded in a larger $\mathfrak{so}(d)$, together with an independent, fully released re-implementation. (5) Five-seed evidence on closure, algebra-type identification and cross-seed structural alignment, including negative results at the originally planned closure weight.

2 Related work

Known-group equivariance assumes the group and its action [2, 1]. **Automatic symmetry discovery** infers generators from predictors [3], adversarially [11], in learned latent spaces [12], beyond affine families

[10], through task gradients [6], or as one-parameter transformations [4]. Our question is narrower: given transition evidence and a generator count, do the learned generators form an approximately closed algebra? **Group-structured transition representations** [8, 7] share our setting. Representation holonomy [9] likewise emphasizes path-dependent structure that pointwise similarity misses. Our alignment experiment (Sec. 6.3) gives a concrete instance of that gap.

3 Problem setting

Observations $x = h(u) \in \mathbb{R}^p$ come from states u through an injective smooth h . We observe triples $(x_0, x_1, x_2) = (h(u), h(g_1 u), h(g_2 g_1 u))$ with unknown near-identity g_1, g_2 from a connected compact group. No group element, coordinate or algebra label is given; K is supplied. We learn an encoder E , decoder D , action inference $q : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^K$ and $A_1, \dots, A_K \in \mathfrak{so}(d)$, with $A(a) = \sum_k a_k A_k$ and $\rho(a) = \exp(A(a)) \in SO(d)$.

4 BracketNet

Local and compositional terms. With $z_t = E(x_t)$ and $a_{st} = q(z_s, z_t)$: $\mathcal{L}_{\text{rec}} = \sum_{t=0}^2 \|D(z_t) - x_t\|^2$, $\mathcal{L}_{\text{loc}} = \sum_{t=0}^1 \|\rho(a_{t,t+1})z_t - z_{t+1}\|^2 + \|D(\rho(a_{t,t+1})z_t) - x_{t+1}\|^2$, and $\mathcal{L}_{\text{comp}} = \|\rho(a_{02}) - \rho(a_{12})\rho(a_{01})\|_F^2$. Coefficients are $0.65 \tanh(\cdot)$, which keeps them in a local chart.

4.1 Lie closure and its invariances

Let $B \in \mathbb{R}^{d^2 \times K}$ hold the vectorized generators and $P_B = B(B^\top B + \varepsilon I)^{-1} B^\top$. Then

$$\mathcal{L}_{\text{close}} = \frac{1}{K^2} \sum_{i,j} \|(I - P_B) \text{vec}[A_i, A_j]\|_2^2. \quad (2)$$

Invariance (corrected). For $\varepsilon = 0$, P_B depends only on the span. The commutators, however, transform quadratically under a change of coordinates $B \mapsto BM$. Hence $\mathcal{L}_{\text{close}}$ is invariant for orthogonal M , scales as s^4 under $M = sI$, and is not invariant for general invertible M . Its *zero set* is $GL(K)$ -invariant. With $\varepsilon > 0$, P_B is only approximately span-dependent. Each generator is normalized to $\|A_k\|_F = \sqrt{2}$, and $\mathcal{L}_{\text{basis}} = \|B^\top B - 2I\|_F^2$ keeps the basis near-orthogonal, which keeps the residual well-conditioned. Because the normalization fixes $\text{diag}(B^\top B) = 2$, $\mathcal{L}_{\text{basis}}$ acts only on off-diagonal entries: targets I and $2I$ give identical gradients (tested numerically). For diagnostics we also report the span-only quantity $\kappa = ((\binom{K}{2})^{-1} \sum_{i < j} \| [U_i, U_j] \|_F^2)^{1/2}$, where U is a $\sqrt{2}$ -scaled orthonormal basis of the span.

4.2 Fit–close–refit

We optimize $\mathcal{L} = 4\mathcal{L}_{\text{rec}} + 32\mathcal{L}_{\text{loc,lat}} + 2\mathcal{L}_{\text{loc,obs}} + 1.2\mathcal{L}_{\text{comp}} + \lambda(t)\mathcal{L}_{\text{close}} + \mathcal{L}_{\text{basis}} + w_{\text{cov}}\|\text{Cov}(z) - I\|_F^2$. $\lambda(t) = 0$ for the first 35% of updates and ramps linearly to λ by 70%. For the final 30% the generators are frozen while E, D, q adapt.

5 Why closure matters

Proposition 1 (Exact closure integrates). *If $A_1, \dots, A_K \in \mathfrak{so}(d)$ are linearly independent and $[A_i, A_j] \in \text{span}(A)$ for all i, j , then $\mathfrak{h} = \text{span}(A)$ is a Lie subalgebra of $\mathfrak{so}(d)$. It integrates to a unique connected immersed Lie subgroup $H \subseteq SO(d)$.*

Lemma 1 (Loss controls the closure constant). *If $B^\top B = 2I$, then $\|(I - P_a)[X, Y]\|_F \leq \delta \|X\|_F \|Y\|_F$ for all $X, Y \in \mathfrak{a}$ with $\delta = \frac{K}{2} \sqrt{\mathcal{L}_{\text{close}}} \ (\varepsilon = 0)$.*

Proposition 2 (Approximate closure controls BCH escape). *Under the hypothesis of Lemma 1, for $\|X\|_F, \|Y\|_F \leq r \leq r_0$ inside the BCH convergence domain, $\text{dist}(\log(e^X e^Y), \mathfrak{a}) \leq \frac{1}{2} \delta r^2 + C(r_0) \delta r^3$.*

Unlike a bound with a δ -independent $C_2 r^3$ term, this one vanishes under exact closure, consistent with Proposition 1.

Proposition 3 (Chained orthogonal stability). *If $z_{t+1} = R_t z_t + e_t$ and $\hat{z}_{t+1} = R_t \hat{z}_t$ with $R_t \in SO(d)$, then $\|\hat{z}_T - z_T\| \leq \|\hat{z}_0 - z_0\| + \sum_{t < T} \|e_t\|$.*

Remark 1 (Closure does not identify the representation). *Closure constrains the algebra, not the embedding. The standard $\mathfrak{so}(3) \subset \mathfrak{so}(4)$ (acting on $\mathbb{R}^3 \oplus \mathbb{R}$) and a chiral $\mathfrak{su}(2) \subset \mathfrak{so}(4)$ are both exactly closed, but with $\sqrt{2}$ -normalized orthonormal bases they give $\kappa = \sqrt{2}$ and $\kappa = 2$ respectively. They are distinguished by the participation ratio $\text{tr}(X^\top X)^2 / \text{tr}((X^\top X)^2)$ of unit span elements (2 vs. 4).*

6 Experiments

6.1 Protocol

Data. $SO(2)$, T^2 and $SO(3)$ act on $\mathbb{R}^d$ with an invariant coordinate ($d = 3, 5, 4$). In particular $SO(3)$ occupies a 3-dimensional subspace of $\mathfrak{so}(4)$, where a generic 3-dimensional span does not close. The mean closure of random normalized 3-spans is 0.67. Local coefficients are $\mathcal{N}(0, 0.24^2)$. Observations use one fixed map per group, $x = Qz + 0.18 \tanh(Bz)$, shared across splits. Each run trains on 1,400 triples and is evaluated on 500 ten-edge sequences. Validation seeds are 5–6 and test seeds 10–14. All methods use 420 updates and identical networks. Unstated details (state distribution, $p = d$, regularizer form and weight, ε , ramp shape) are listed in the released `docs/ASSUMPTIONS.md`.

Hyperparameters selected on validation only. With $w_{\text{cov}} = 1$ the latent collapsed (4 of 5 covariance eigenvalues below 0.1). We therefore chose $w_{\text{cov}} = 16$ from $\{1, 4, 16, 64\}$ by a method-agnostic criterion: 1-step MSE of the two baselines, with closure not consulted. *Protocol A* keeps the planned $\lambda = 0.1$. On validation it left closure unchanged (Fig. 4), because the effective scale of λ depends on loss-reduction conventions. *Protocol B* extends the grid to $\{0.05, \dots, 30\}$. It picks the λ with lowest mean validation closure whose mean validation 10-step MSE does not exceed that of +Comp. ($\lambda = 30$). This rule was fixed after inspecting validation results and before any test run; we report both protocols.

Baselines and metrics. LOCAL: reconstruction and one-step transport. +COMP.: adds $\mathcal{L}_{\text{comp}}$. BRACKETNET: adds closure and the schedule. These are controlled objective ablations, not reimplementations of LieGAN/LaLiGAN. Chained k -step MSE composes actions inferred from each observed adjacent pair, so it measures consistency, not forecasting. We report means $\pm$ sample s.d. over seeds, paired t intervals, and exact sign-flip tests. With $n = 5$ the smallest attainable two-sided p is 0.0625.

6.2 Main results

Closure at the planned weight: no effect. Under Protocol A, BracketNet’s mean closure was higher than +Comp.’s on both groups: T^2 0.315 vs. 0.212, lower on 0/5 seeds; $SO(3)$ 0.687 vs. 0.585, lower on 2/5 seeds (Table 1). The 74–75% reductions reported in an earlier draft are therefore *not reproduced* at $\lambda = 0.1$.

Closure at the validation-selected weight: large, consistent reduction. Under Protocol B, mean closure fell from 0.212 to 0.0023 on T^2 (−98.9%) and from 0.585 to 0.0082 on $SO(3)$ (−98.6%), lower on 5/5 seeds in each group. Baseline closures are bimodal across seeds, so paired t intervals are wide: $SO(3)$ -0.577 ± 0.560 excludes zero narrowly, while T^2 -0.209 ± 0.403 does not. The sign-flip $p = 0.0625$ for both is the floor for five seeds.

Algebra type. A threshold of 0.918 was fixed midway between the pooled validation ranges (T^2 : 0.016–0.784; $SO(3)$: 1.052–1.425). It classified 10/10 Protocol-B test runs correctly, against 9/10 for Local, +Comp. and Protocol A (Fig. 2). The single errors are T^2 runs with spurious non-commutativity. Protocol B’s $SO(3)$ commutator norms (1.428 ± 0.019) concentrate at the true value $\sqrt{2}$; the baselines spread over 1.13–1.59.

Transport: no significant difference. Relative to +Comp., Protocol B’s 10-step MSE changed by +5.1% on T^2 ($+0.0084 \pm 0.0277$) and −16.9% on $SO(3)$ (-0.0285 ± 0.0363). Both intervals include zero (Fig. 5). We make no transport claim.

Table 1: Held-out results (seeds 10–14, mean $\pm$ s.d.). Closure is identically zero for one-generator $\text{SO}(2)$. Absolute MSE values depend on the data scale and are not comparable with earlier drafts.

Group	Method	1-step MSE	10-step MSE	Closure residual	Commutator norm
$\text{SO}(2)$	Local	0.0055 ± 0.0005	0.0535 ± 0.0122	0.0000 ± 0.0000	0.000 ± 0.000
$\text{SO}(2)$	+Comp.	0.0056 ± 0.0007	0.0537 ± 0.0112	0.0000 ± 0.0000	0.000 ± 0.000
$\text{SO}(2)$	BracketNet (A, $\lambda=0.1$)	0.0056 ± 0.0007	0.0539 ± 0.0111	0.0000 ± 0.0000	0.000 ± 0.000
$\text{SO}(2)$	BracketNet (B, $\lambda=30$)	0.0056 ± 0.0007	0.0539 ± 0.0111	0.0000 ± 0.0000	0.000 ± 0.000
T^2	Local	0.0177 ± 0.0039	0.1648 ± 0.0307	0.2186 ± 0.3220	0.505 ± 0.478
T^2	+Comp.	0.0174 ± 0.0042	0.1640 ± 0.0324	0.2116 ± 0.3283	0.475 ± 0.497
T^2	BracketNet (A, $\lambda=0.1$)	0.0183 ± 0.0040	0.1675 ± 0.0319	0.3148 ± 0.3091	0.702 ± 0.413
T^2	BracketNet (B, $\lambda=30$)	0.0179 ± 0.0050	0.1724 ± 0.0434	0.0023 ± 0.0038	0.051 ± 0.051
$\text{SO}(3)$	Local	0.0180 ± 0.0067	0.1567 ± 0.0287	0.5928 ± 0.4674	1.369 ± 0.168
$\text{SO}(3)$	+Comp.	0.0170 ± 0.0066	0.1682 ± 0.0354	0.5851 ± 0.4593	1.374 ± 0.171
$\text{SO}(3)$	BracketNet (A, $\lambda=0.1$)	0.0181 ± 0.0054	0.1607 ± 0.0286	0.6874 ± 0.3955	1.324 ± 0.164
$\text{SO}(3)$	BracketNet (B, $\lambda=30$)	0.0194 ± 0.0111	0.1397 ± 0.0363	0.0082 ± 0.0112	1.428 ± 0.019

6.3 Structural alignment between independently trained representations

Latent representations are identifiable at best up to conjugacy. We therefore ask whether models trained from different seeds learn the same *algebra* once their latents are aligned. All models share the observation map, so we encode a common probe set (200 fresh sequences), align each pair of seeds with orthogonal Procrustes R , and measure (i) linear CKA of the latents and (ii) the mean squared sine of principal angles between $\text{span}(RA^{(a)}R^\top)$ and $\text{span}(A^{(b)})$. We score each model against the ground truth the same way. On $\text{SO}(3)$, Protocol-B models agreed with each other at 0.034 ± 0.021 and with the truth at 0.019. Local, +Comp. and Protocol A gave 0.42–0.45 between seeds, close to the random-span level of 0.50 (Fig. 3). On T^2 the corresponding numbers were 0.23 vs. 0.34–0.42 (chance 0.80). Linear CKA was essentially identical across methods (0.892–0.897 on $\text{SO}(3)$, 0.965–0.967 on T^2). Pointwise similarity cannot reveal whether two representations share their transformation algebra. A caveat: inferred *actions* remained poorly aligned for every method (normalized disagreement with the truth ≥ 0.51), so recovering the algebra did not imply accurate action inference within the training budget.

6.4 Ablations and failure analysis

These ablations were run post hoc on the test seeds and were not used for selection. (1) *Vacuous ambient*. With $\text{SO}(3)$ in $\mathfrak{so}(3)$, closure is exactly zero for every method and seed. (2) *Split-dependent map*. Sampling a separate test map leaves closure unchanged but raises +Comp.’s 10-step MSE by 0.20 (T^2) and 0.36 ($\text{SO}(3)$); it measures domain shift, not symmetry recovery. (3) *Gram target I*. Identical to target $2I$ up to 4×10^{-8} in closure, as predicted in Sec. 4.1. This pilot cannot explain any earlier failure. (4) *Constant penalty* ($\lambda = 30$ throughout, no freeze). Similar closure but worse mean transport ($T^2 + 0.030 \pm 0.056$, $\text{SO}(3) + 0.075 \pm 0.125$). The direction favours staging, but the intervals include zero. (5) *Budget*. At 2,000 updates every method fits transport better, and baseline closure on T^2 improves (0.052). On $\text{SO}(3)$ baseline closure stays high (0.64–0.66), while Protocol B reaches 1.4×10^{-4} (Fig. 6). Better transition fit does not by itself produce closure for the non-abelian group. In that setting, one Protocol-B $\text{SO}(3)$ seed converged to an exactly closed chiral $\mathfrak{su}(2)$ (commutator norm 2.000, participation ratio 4.00), illustrating Remark 1.

7 Limitations

Weight scale and budget. The useful closure weight depended on loss conventions; we needed $\lambda = 30$ where an earlier plan used 0.1. At 420 updates no method had converged, and the freeze stage halts generator learning before convergence. **Statistics.** Five seeds bound exact p -values at 0.0625, and bimodal baselines widen t intervals. **Scope.** Known K ; compact orthogonal actions only; synthetic data; endpoint-inferred

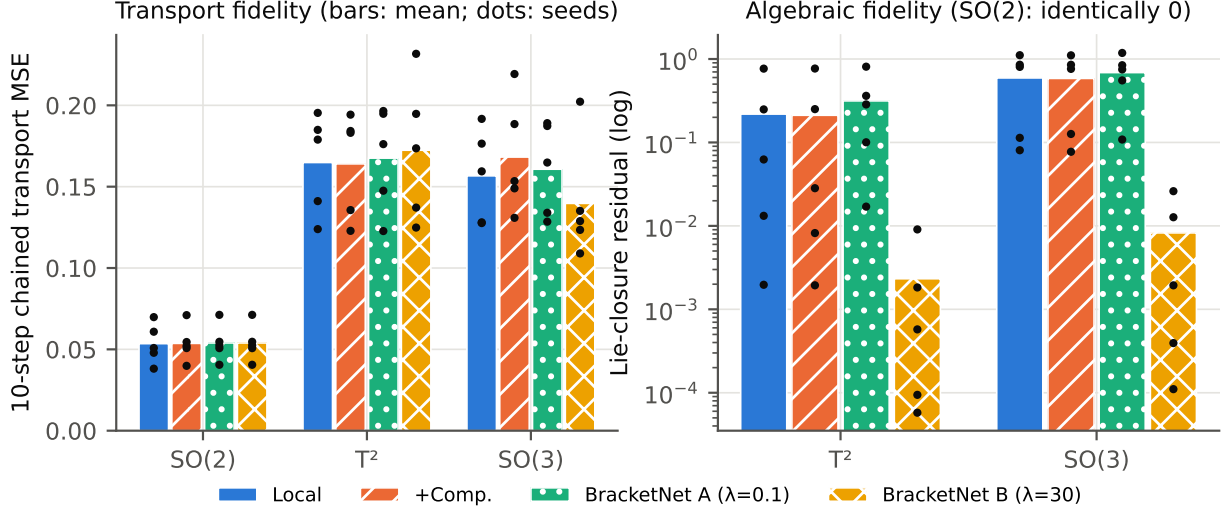


Figure 1: Left: 10-step chained transport. Right: closure residual (log). Bars: means; dots: seeds.

chained transport. **Identifiability.** Closure does not fix the representation (chiral vs. standard), and latents are aligned only up to conjugacy.

8 Conclusion

A basis-normalized commutator residual combined with fit–close–refit training can drive learned generator spans to near-exact Lie subalgebras. In our experiments this made algebra type identifiable and made independently trained models agree on their algebra, which representation-similarity measures such as CKA do not detect. These gains required re-selecting the closure weight on validation data: the originally planned weight had no effect. We found no significant transport advantage.

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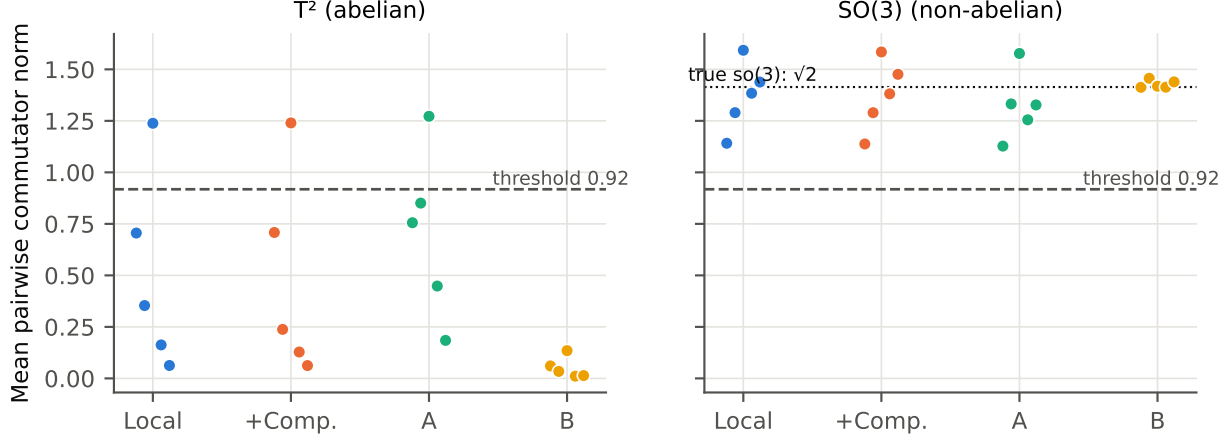


Figure 2: Commutator norms of held-out runs. Dashed: threshold fixed on validation. Dotted: true $\mathfrak{so}(3)$ value.

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A Proofs

Proposition 1. The commutator is bilinear, antisymmetric and satisfies Jacobi in any associative algebra. Closure makes $\mathfrak{h}$ a Lie subalgebra, and the Lie subgroup theorem [5] gives the unique connected immersed subgroup generated by $\exp(\mathfrak{h})$. $\square$

Lemma 1. Write $X = \sum_i x_i A_i$ and $Y = \sum_j y_j A_j$, so that $\|X\|_F = \sqrt{2}\|x\|$ and $\|Y\|_F = \sqrt{2}\|y\|$. Let $r_{ij} = (I - P)[A_i, A_j]$. By Cauchy–Schwarz over the K^2 pairs, $\|\sum x_i y_j r_{ij}\| \leq \|x\| \|y\| (\sum \|r_{ij}\|^2)^{1/2} = \|x\| \|y\| K \sqrt{\mathcal{L}_{\text{close}}} = \frac{K}{2} \sqrt{\mathcal{L}_{\text{close}}} \|X\|_F \|Y\|_F$. $\square$

Proposition 2. In Dynkin form, the degree- n BCH term is a combination of right-nested commutators $Z_n = [W_1, [W_2, \dots [W_{n-1}, W_n]]]$ with $W_i \in \{X, Y\}$. Let $N = I - P_{\mathfrak{a}}$ and $e_n = \sup \|N Z_n\|$. Then $e_1 = 0$ and $e_2 \leq \delta r^2$. Writing $Z_n = [W, P Z_{n-1}] + [W, N Z_{n-1}]$, we get $e_n \leq \delta r \|Z_{n-1}\| + 2r e_{n-1} \leq \delta 2^{n-2} r^n + 2r e_{n-1}$, hence $e_n \leq n 2^{n-2} \delta r^n$. The degree-1 terms lie in $\mathfrak{a}$ and the degree-2 term is $\frac{1}{2}[X, Y]$. Summing the remaining terms against the absolutely convergent BCH coefficients for $r \leq r_0$ gives the bound, with $C(r_0)$ collecting the series tail. $\square$

Proposition 3. $d_{t+1} = R_t d_t - e_t$ and $\|R_t d_t\| = \|d_t\|$; iterate. $\square$

B Implementation details

E , D and q are 2-hidden-layer SiLU MLPs of width 64. Optimization: AdamW (lr 2×10^{-3} , weight decay 10^{-5}), batch 160, gradient clipping 5, 420 updates. $\varepsilon = 10^{-6}$, $w_{\text{basis}} = 1$, $w_{\text{cov}} = 16$. Runs use one CPU thread and take 2–14s each (420 updates, 4 concurrent workers). Re-running stored runs reproduced every checked metric bit for bit on the same machine.

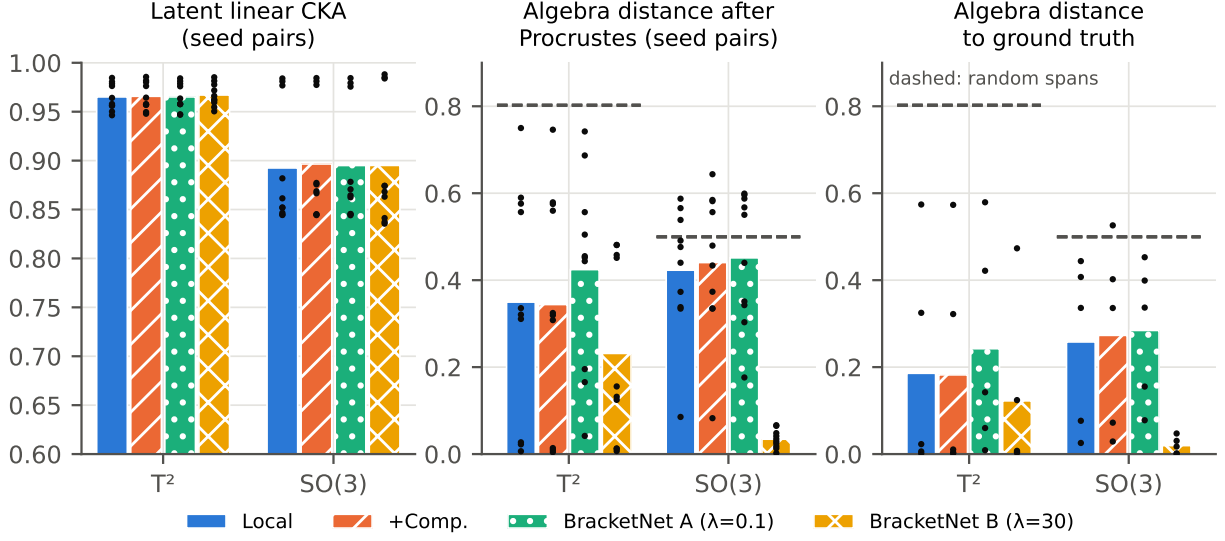


Figure 3: Cross-seed alignment (10 seed pairs) and alignment to ground truth (5 seeds). CKA cannot distinguish the methods; algebra distance after Procrustes can.

C Additional results

D Reproducibility

The repository contains the model and data-generation code, per-run JSON results for all 255 training runs, the locked-hyperparameter record, figure and table scripts, the mathematical unit tests, and `run_all.sh`, which reproduces every number in this paper.

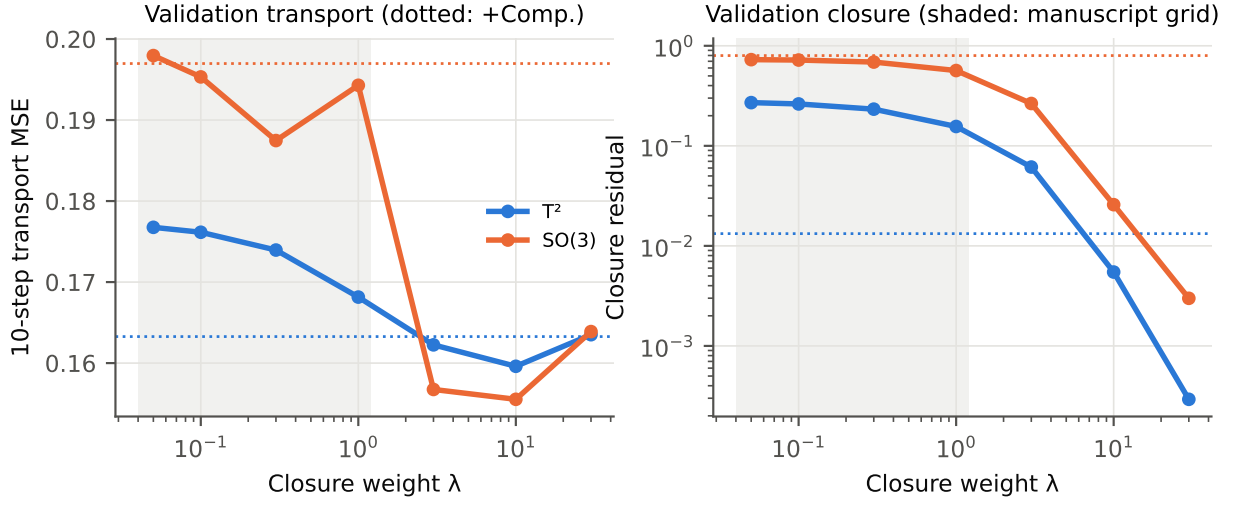


Figure 4: Validation sweep (seeds 5–6). The originally planned grid (shaded) does not reduce closure below +Comp. (dotted); $\lambda \geq 10$ does, without degrading validation transport.

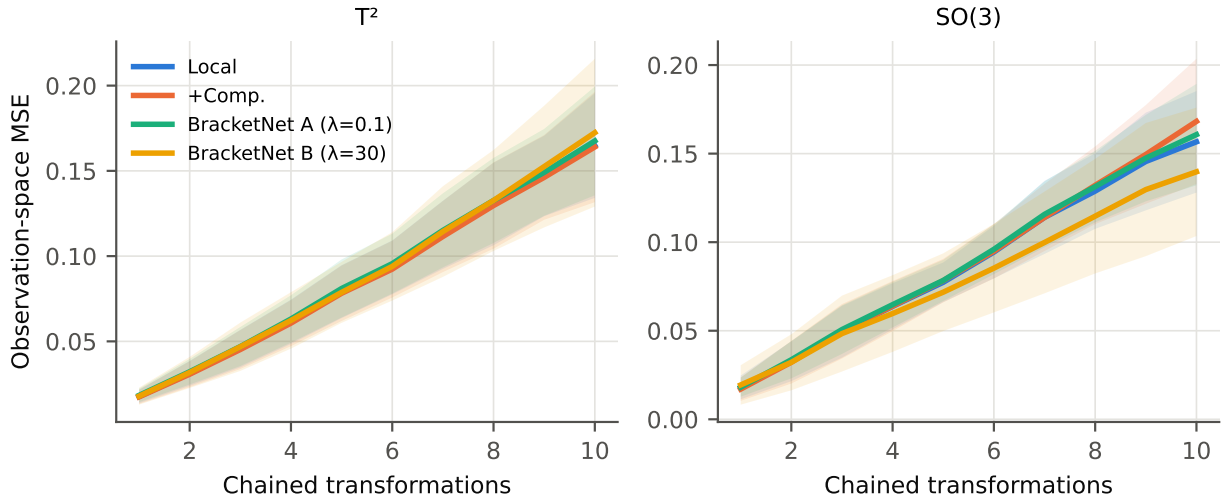


Figure 5: Chained transport error vs. path length; shading is one s.d. over five seeds.

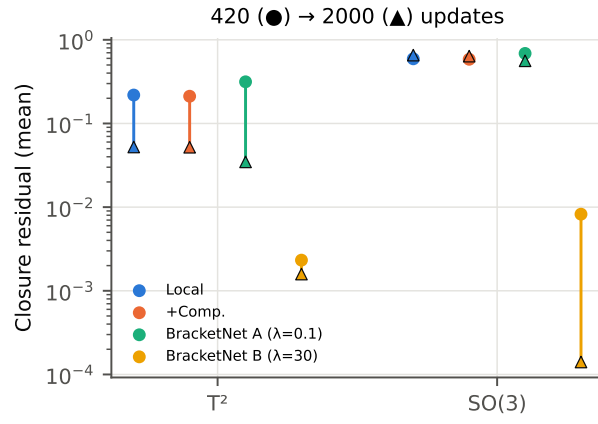


Figure 6: Mean closure at 420 vs. 2,000 updates (post-hoc ablation, test seeds).